

Fady Youssef
Kyle Connall
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The relationship between Sleep and Caffeine usage

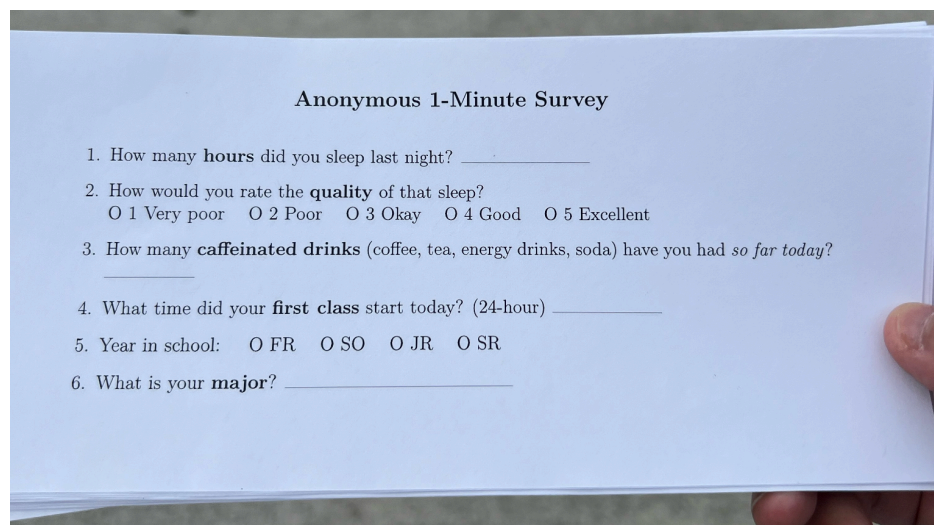
A. Introduction

In our study we asked the question, “Do Gonzaga students who sleep less, drink more caffeine in the morning, especially when their first class is early?” Our population for this study was Gonzaga students that passed by our booth, from 12pm to 2 pm behind College hall on 4/19/25.

We based our study on two articles from the National Library of Medicine: *Subjective Sleep Quality, Caffeine, and Dieting Behaviors Among University-Attending Young Adults*, and *Early Morning University Classes Are Associated with Impaired Sleep and Academic Performance*. These articles talk about how 8am classes make students wake up earlier and affect their academic performance, through attendance and cognitive functions. These articles also showed that increasing caffeine intake and decreasing sleep quality, and when combining both an early morning class and caffeine intake, can lead to even worse sleep.

B. Methods

Our experiment was an observational study rather than a controlled experiment. We allowed volunteer contestants to come to the booth and flip a coin, if the coin landed on heads the contestant was able to participate in the study, if the coin landed on tails, the participant was turned away with a stress ball and candy as a consolation prize. Each validated contestant was then given a survey and asked to fill out to the best of their knowledge. We tried to limit the amount of intervention we had with the survey, to unbiased the results. The survey that each contestant took was then collected by the proctors (Fady and Kyle). Below all sections of the paper is the survey that was given out.



Anonymous 1-Minute Survey

1. How many **hours** did you sleep last night? _____
2. How would you rate the **quality** of that sleep?
O 1 Very poor O 2 Poor O 3 Okay O 4 Good O 5 Excellent
3. How many **caffeinated drinks** (coffee, tea, energy drinks, soda) have you had *so far today*?

4. What time did your **first class** start today? (24-hour) _____
5. Year in school: O FR O SO O JR O SR
6. What is your **major**? _____

Figure 1: Survey given to Gonzaga students

C. Data

Our study was based on an observational survey with 42 students. We didn't assign anyone to groups or apply treatments, we just recorded responses as they came in. The data showed a general trend that students who slept less tended to drink more caffeine. Most students who got around 6 to 8 hours of sleep reported drinking between 0 and 2 cups. Students who slept under 6 hours showed more variation, with some drinking up to 4 or 5 cups. There were also students who didn't drink any caffeine regardless of how much they slept, which shows how different individual habits can be.

There were a few issues with how we measured caffeine. We asked students how many cups they had, but not how much caffeine was in those drinks. That means one person could have four weak cups of coffee, and another could have one strong energy drink, and they'd both count as the same in the data even though the actual caffeine is different. Some students might've also guessed or estimated instead of being exact, especially if they didn't really track how much they drank. It's also possible that students underreported caffeine or sleep to seem healthier or more put together. On top of that, we collected data at a specific time of day, so people who were rushing between classes or avoiding the courtyard altogether weren't represented. Even though 42 students gave us enough data to run a proper analysis, in the grand scheme of over 8,000 students at Gonzaga, a bigger and more complete sample could've told us a lot more. We also didn't ask about other important things like workload, stress levels, or sleep routines, all of which could've played a big role in both caffeine use and sleep habits. All of these things could've introduced some bias or noise into the results.

D. Analysis

Null Hypothesis h_0 : There is no linear relationship between hours of sleep and caffeine consumption in the population.

Alternative Hypothesis h_a : There is a linear relationship between hours of sleep and caffeine consumption in the population.

We ran linear regression in R using the exact code below:

```
####Authours:
```

```
#Kyle Connall
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#Fady Youssef
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```
df <- sleepCaffeine_Data
```

```
#Run the linear regression
```

```
model <- lm(caffeineDrinks ~ hoursSlept, data = df)
```

```
summary(model)
```

```
#Plot data and regression line
```

```
plot(df$hoursSlept, df$caffeineDrinks,  
     main = "Caffeine Consumption vs. Hours of Sleep",
```

```

xlab = "Hours of Sleep",
ylab = "Caffeinated Drinks",
pch = 19, col = "blue")
abline(model, col = "red", lwd = 2)
#display correlation coefficient (r)
r_squared <- summary(model)$r.squared
r <- sqrt(r_squared)
r

b <- coef(summary(model))["hoursSlept", "Estimate"]
se <- coef(summary(model))["hoursSlept", "Std. Error"]
t_val <- qt(0.975, df = nrow(df) - 2)
lower <- b - t_val * se
upper <- b + t_val * se
c(lower, upper)
To produce:

```

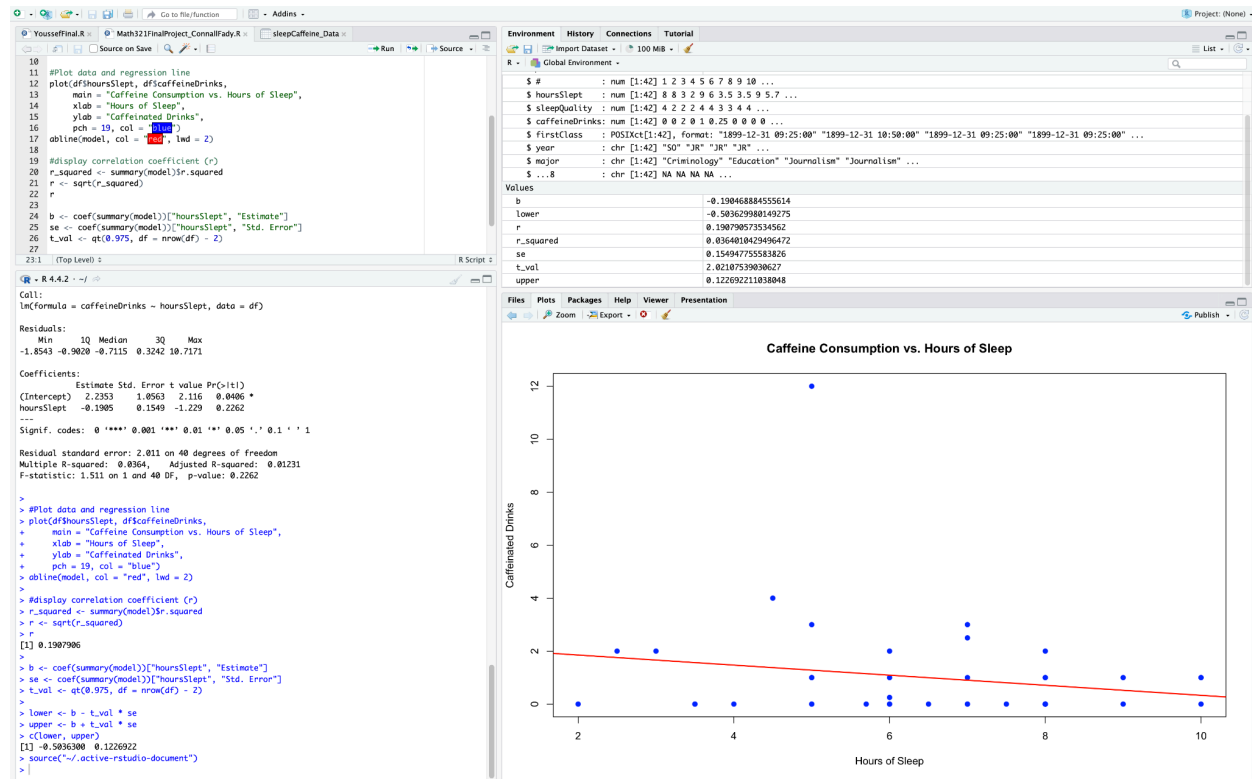


Figure 2: Scatterplot of Caffeinated Drinks vs. Hours of Sleep with Regression Line

From the summary we are given:

T-val for hoursSlept: -1.226

P-val: .2262

r^2 : .0364

corr (r): .1907

And since the $p\text{-value} > .05$, we fail to reject the null hypothesis indicating the data doesn't provide enough statistically significant evidence between hours of sleep and caffeine consumption.

We also calculated the 95% confidence interval for the slope of hoursSlept which resulted in a range from $-.05836$ to $.1227$, including zero. These results support the slope not being statistically significant and showing no strong linear relationship[between sleep and caffeine consumption in our samples.

E. Conclusion

Our results didn't show a strong enough link between hours of sleep and caffeine consumption to be considered statistically significant. The p-value was 0.2262 , and the 95 percent confidence interval for the slope ranged from -0.5836 to 0.1227 , which includes zero. That means we can't say for sure there's a real connection. The r-squared was just 0.0364 , so sleep only explained about 3.6 percent of the differences in how much caffeine people drank. Still, we did notice a loose pattern where students who slept less sometimes drank more, but it wasn't consistent enough to count. This kind of research matters because one of the studies we cited showed how early classes and caffeine can combine to hurt sleep and academic performance. So even if our results weren't conclusive, the question is still important.

If we were to do this again, we'd want a larger and more balanced sample that better represents all types of students at Gonzaga. We'd also change how we collected the data. Instead of asking for caffeine in cups, we'd ask for the type of drink and how much caffeine it actually had in milligrams. We'd also include more questions about sleep quality, stress, and workload, since those could all affect both sleep and caffeine use. Collecting data throughout the day, not just in one time block, would help avoid time-based bias too. With all of that, we'd have a much better shot at finding real trends and understanding what actually drives caffeine habits on campus.

References

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